

Robotic Arm Grab

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1. Experiment Purpose
2. Experiment Steps
3. Experiment Source Code
4. Program Download
5. Experiment Result

1. Experiment Purpose

Lastbit is equipped with a 3-axis robotic arm. In the previous lesson, we learned how to make the robotic arm perform a dance-like effect by controlling the three joint servos of the robotic arm at the same time. In this section, we will try to control a single joint servo of the robotic arm to grab a block.

2. Experiment Steps

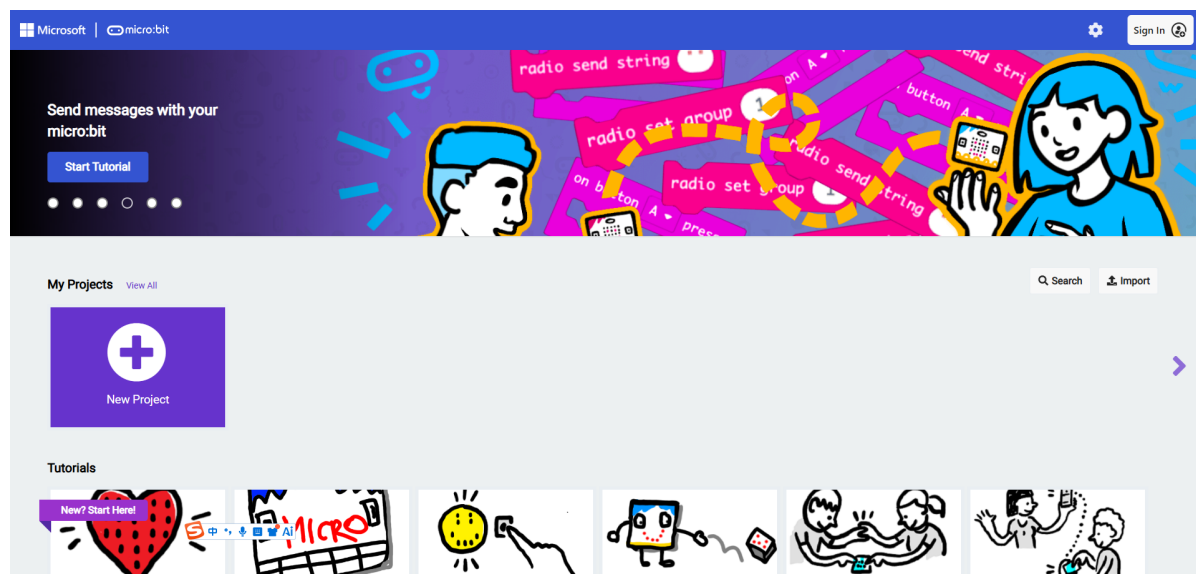
S1	Gripper joint of the robotic arm
S2	Vertical degree-of-freedom joint of the robotic arm
S3	Horizontal degree-of-freedom joint of the robotic arm

1. Before the experiment, use a Micro USB data cable to connect the Micro:bit main board to the computer, and the computer will display the MICROBIT drive.



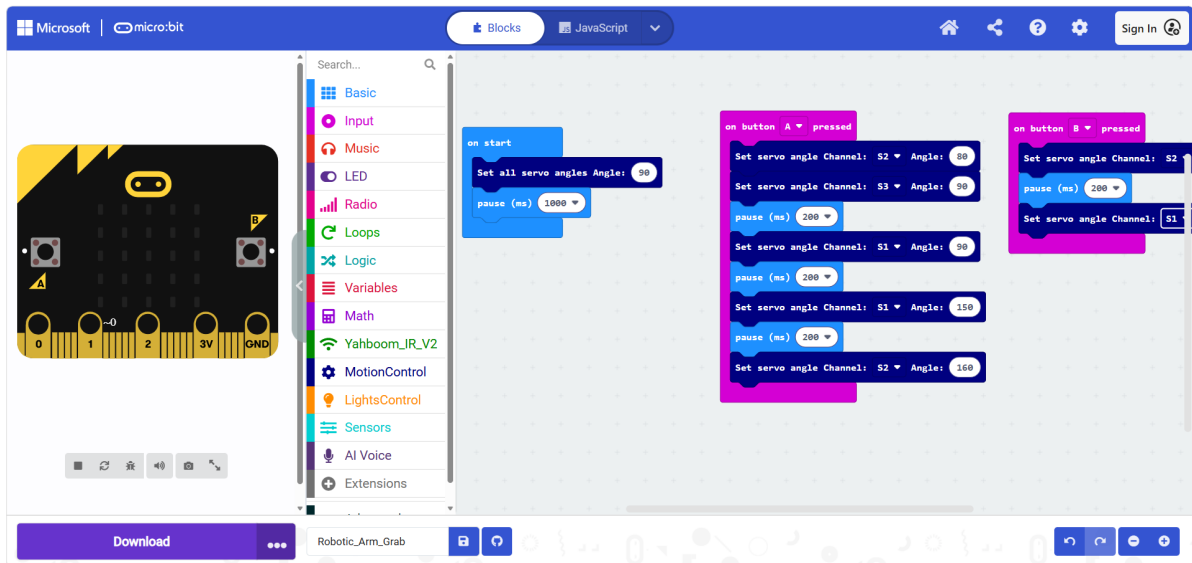
2. Open MakeCode through the link below.

[Microsoft MakeCode for micro:bit](#)

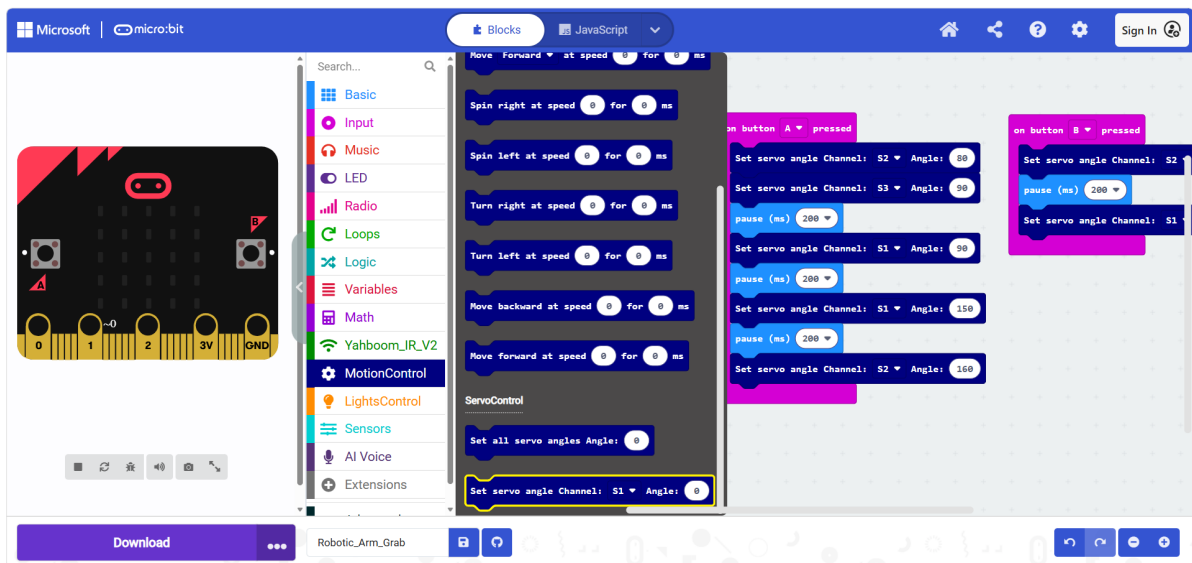


3. Drag `5.Robotic_Arm_Grab.hex` to the New Project page, and MakeCode will automatically import and open the project.

3. Experiment Source Code

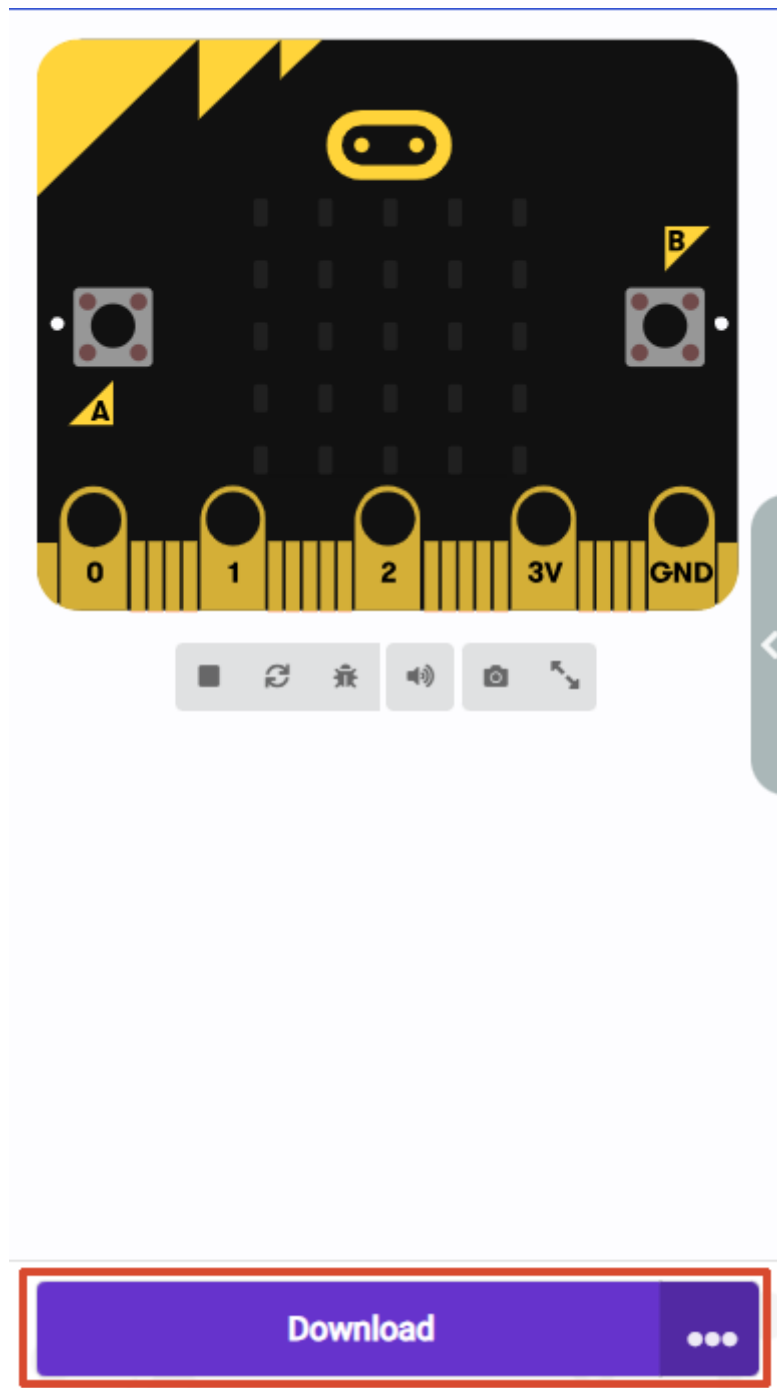


The blocks in this chapter belong to the `Servo Control` category under `Lastbit Motion Control`.



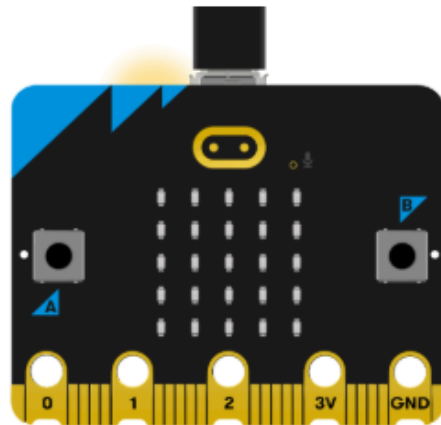
4. Program Download

1. Click Download in the lower left corner to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Micro:bit, the system will first prompt you to connect the device. If it is not connected, connect the data cable to the Micro:bit first; if it is already connected, click [Next].

1. Connect your micro:bit to your computer

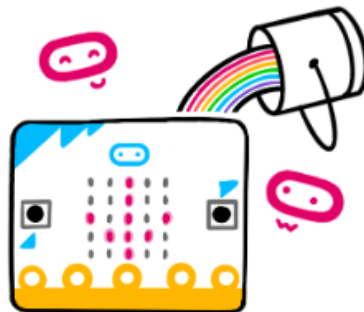


 Next

3. If the Micro:bit is detected to be connected normally, the following interface will be displayed, and you can click [Download].

 **Connected to micro:bit** 

Your micro:bit is connected! Pressing 'Download' will now automatically copy your code to your micro:bit.



 Download

If the Micro:bit data cable is not properly connected or multiple Micro:bits are connected, the system will ask you to manually select the device. You can choose one of the following two methods:

- [Download as File]
- [Pair]

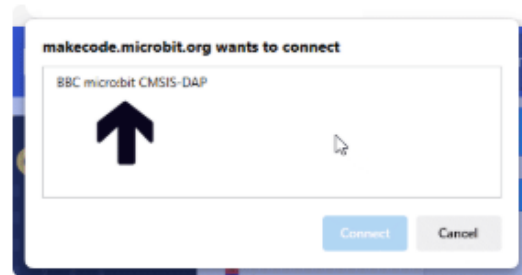
2. Pair your micro:bit to your browser



Press the Pair button below.

A window will appear in the top of your browser.

Select the micro:bit device and click Connect.

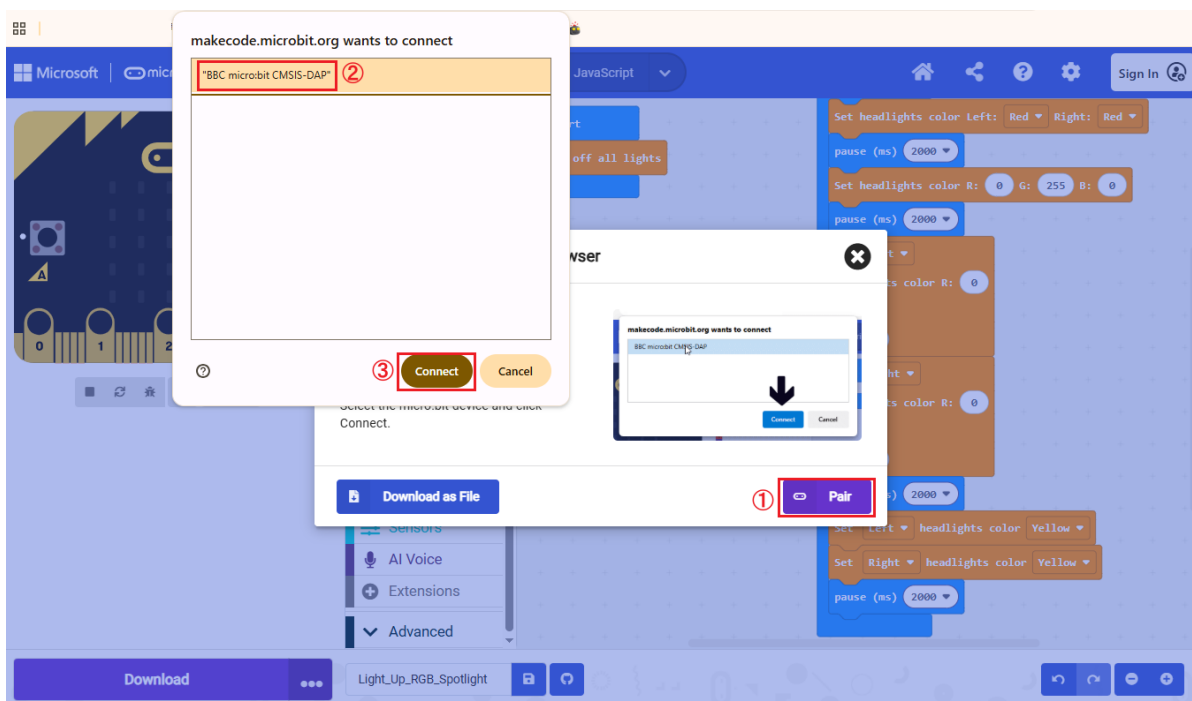


Download as File

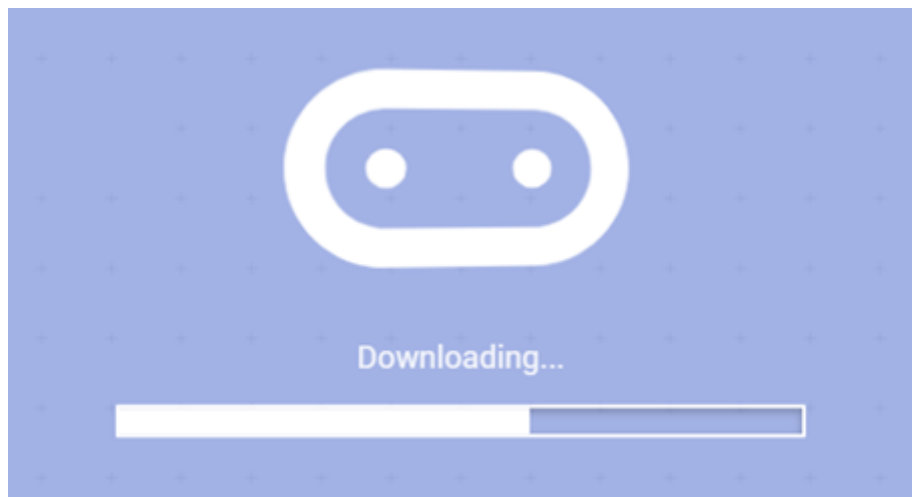
Pair

When selecting [Download as File], the system will download the currently open program as `microbit-<project name>.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive to complete the download.

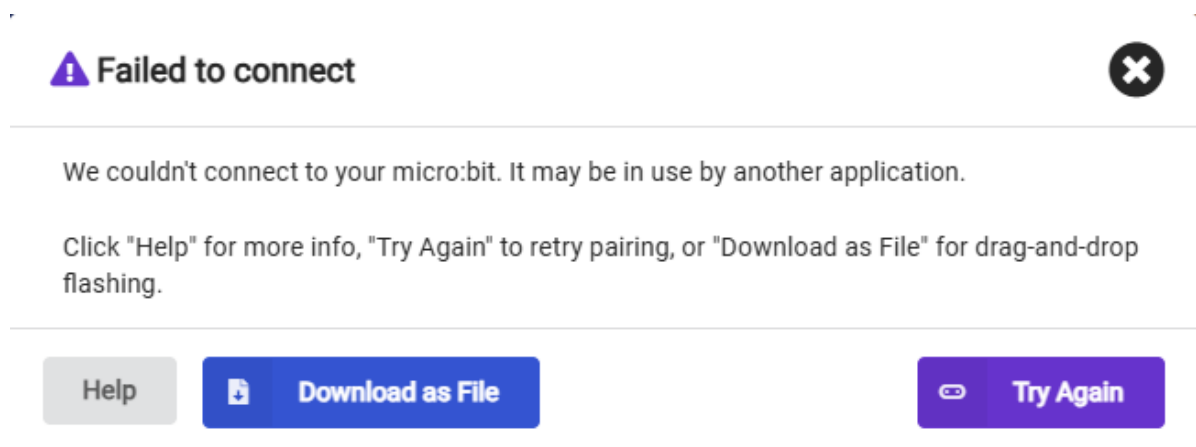
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC micro:bit CMSIS:DAP`.



4. The following interface indicates that the program is being downloaded. Wait for the progress bar to finish to download the program to the Micro:bit.

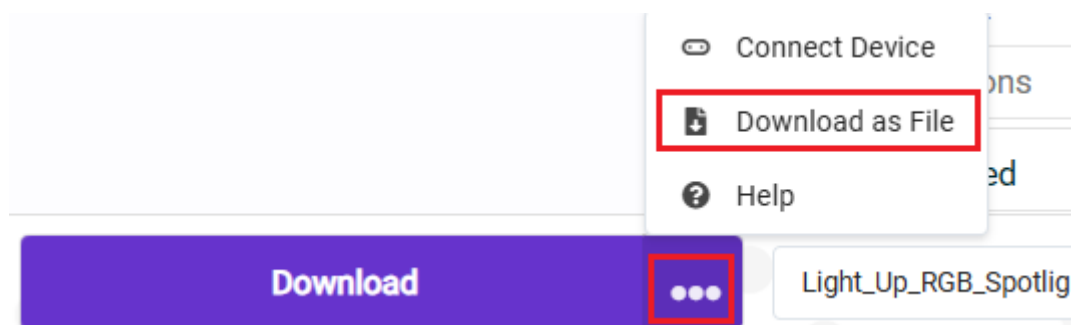


If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.

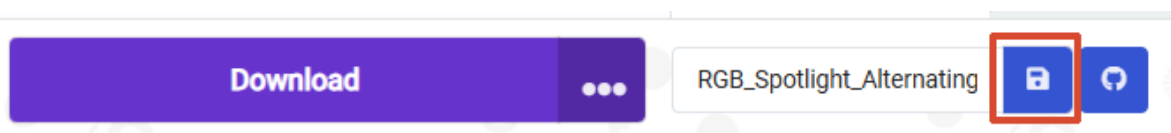


If you want to save the file directly, you can use the following two methods:

Method 1: Click [...] next to Download, and then select [Download as File].



Method 2: Click [Save] after the project name editing field.



5. Experiment Result

After flashing, turn on the power switch (toggle the switch to the ON state), and the robotic arm will first return to its initial state. Press the A button to grab and lift the object, and press the B button to put down the object and release the gripper.

